Thesis changes since the 24 October version: before and after, for your review

Prepared 25 October 2026. **BEFORE** is the text you sent Prof. Kong on 24 October, with her Chapter 1 edits already ported in (commit `a1f1af2`). **AFTER** is the text now (the 25 October YZU file and the other copies are built from it). Every change below is already applied and committed, so each one can be reverted, but none has been shown to you before; this document is that preview. Nothing here has been sent to her.

A. Her comments, and what became of each

Her item	What was done	Where to look	Your call
Overall: the empirical design does not test the hypotheses well enough	Direct tests added: revenue growth on transaction value growth for H1; OLS of the wedge and the ecosystem ratio on platform and home-market features	3.3, 5.2, 5.3; Tables 5.3 and 5.4	Read 5.2 and 5.3. All of it is exploratory
Overall: tighten, get to the point	**Not done.** The main text grew from 21,358 to 23,428 words	see the notes in section D	Choose the cuts
Overall: strengthen the economic implications	Investors, sellers and drivers, and policy, each in a paragraph in 5.3; investors in 7.2	5.3, 7.2	Check the wording on market power
1. Drop the Symbols page; spell out abbreviations	Done. Symbols are in Appendix A; BPS, UNCTAD, OECD, LSEG, QRIS spelled out at first use	Appendix A; 1.1, 1.2, 4.5, 5.10	none
2. Table 1 (literature gaps)	Dropped; the gaps are in the list in 2.4	2.4	none
2. Tables 2 and 7 to Appendix A	Done (terms in 3.1 became prose; sources are Appendix A.2)	3.1, Appendix A	none
2. Tables 3 and 4	Dropped (for the defence slides)	3.4, 3.6	none
2. Table 5 to the appendix	Done (Appendix B.2)	Appendix B.2	none
2. Table 6 as a footnote	Done: the first footnote in the thesis	4.6	none
2. Table 8	Dropped; each section states its label	5 (opening)	none
2. Table 9 kept and cited in 3.2	Done	3.2, 5.1	none
2. Table 10 (test of H1) replaced by the correlation matrix	Done. The group test of take-rate moves is now text and Appendix C	5.2, Table 5.2	none
2. Tables 11 and 12 replaced by summary statistics and OLS	Table 11 replaced (the seven windows moved to Appendix B.3). **Table 12 (the 2023 rise in the count) is still in the text, in 5.7**	5.3, 5.7	Needs her answer
2. Figure 2 as Panels A, B and C	Done. The right panel is now a simple percentage change	Figure 5.1	Check the figure
2. Section 3.3	Replaced by direct tests	3.3	none
3. Appendix A, definition of variables	Done, in her three-column form, with the regression variables in Panel E	Appendix A	none
4. Summary statistics and correlations in one table	Done. Panel B has nine variables; inflation and household spending were left out for width (they correlate 0.80 with the policy rate and 0.92 with GDP growth)	Table 5.2	Say if she wants all eleven

Her item	What was done	Where to look	Your call
4. Correlation of revenue growth and transaction value growth for H1	Done: 0.90 abroad, 0.31 in Indonesia and the region	5.2	none
4. OLS of ln(wedge) or the ecosystem ratio	Done with her controls; wedge in US dollars; size by total assets; take rate not a regressor; errors clustered by platform. Competitive entry and regulatory events are replaced by year dummies	Table 5.4	Needs her answer on events
4. Simple percentage change instead of log change	Done, except the registered market-value test (kept in logs as registered) and Appendix B.3 and Figure 5.2 (an exact split needs logs)	3.2, 3.3, 5.2, 5.6, 5.7	Needs her answer
5. Economic implications in results and conclusion	Done	5.3, 7.2	see above
6. Figures and tables after the References; "[Insert Table N about here]"	Done in the YZU Word file: six figures, then seven tables, each on its own page after the References, with 13 markers placed after the paragraph that first mentions each in its chapter; the Appendices follow, as in her sample. The reading, compact, condensed and editing copies keep them inline	YZU file, after the References	Say if she also wants the markers in the reading copy
6. Captions that explain how to read each table	Done for all seven tables (a "This table..." paragraph under each title)	5.1 to 5.5, 6.1, 6.2	none
23. Abbreviations spelled out at first use	Also fixed on 25 Oct: OLS expanded at first use (3.3); "SEC filings" now read "filings with the US Securities and Exchange Commission"	3.3, 4.4, 5.3	none
In her file: comments 23 (abbreviations), 26 (coverage)	Done	1.1	none
In her file: comment 46 (wedge against transaction value)	Done: it is 0.9994 in levels by construction, so the test is in growth rates	5.2	none
In her file: comment 47 (regression for H2), comments 48 and 49 (entry test inside the further analyses)	**Not done.** No unit for the H2 regression; the entry test is still its own section	3.5, 5.6	Needs her answer

B. Size and numbers

	Before (24 Oct)	After (25 Oct)
YZU Word file, pages	84	100
Reading copy, pages	60	69
Compact copy, pages	27	29
Condensed Word file, pages	26	28
Tables in the YZU file	14	7
Figures	6	6
Main text, words (Chapters 1 to 7)	21,358	23,428
Equations	8	9
Appendices	A to D	A to E (A is the variable definitions)
Automatic checks that pass	314	398
H1: size of take-rate moves, Indonesia against abroad	"about three times" (0.20 against 0.06 in logs)	3.6 times (22 against 6 percent a year, simple change); the log version still gives 0.20 against 0.06

	Before (24 Oct)	After (25 Oct)
H1 in growth rates	not reported	slope 0.99 abroad, 0.40 in Indonesia and the region (p = 0.02 against 1)
H2	“more businesses account for 71 percent of the growth”	the number of businesses grew 27 percent and sales per business 10 percent
H2 read as wider coverage	“42 to 54 percent”	count +12 to 17 percent against sales per business +14 to 17 percent

C. How to read the changes

Each block below is one paragraph of the text. **BEFORE** is the old wording. **AFTER** is the new wording, with ~~deleted words struck through~~ and **added words in bold**. **NEW** and **REMOVED** are paragraphs that exist on one side only. Tables are abbreviated to their first rows. Appendices are not compared line by line: Appendix A (variables), B.5 (classification of the platforms) and C.3 (checks on the new tests) are new; the old Appendices A to D are now B to E.

Abstract

Abstract (opening)

BEFORE

> *In 2023, three major platforms kept about US$3 billion of the US$43 billion that passed through them. Indonesian and regional platforms changed the share of each sale they kept about three times as much as 27 platforms abroad, mostly between 2022 and 2024. So the hypothesis that revenue grows in proportion to the commerce is rejected.*

AFTER (~~deleted~~, added)

> *In 2023, three major platforms kept about US$3 billion of the US$43 billion that passed through them. ~~Indonesian~~* **Abroad, revenue grew in proportion to transaction value. In Indonesia** *and ~~regional~~* **the region it did not:** *platforms changed the share of each sale they kept ~~about three~~* **3.6** *times as much as 27 platforms abroad, mostly between 2022 and ~~2024. So the~~* **2024, so that** *hypothesis ~~that revenue grows in proportion to the commerce~~ is rejected.*

Chapter 1. Introduction

Chapter 1. Introduction (opening)

BEFORE

> *Over the past decade, digital commerce has changed how goods and services are bought and sold. A growing part of it passes through platforms, such as online marketplaces and ride-hailing and delivery apps. Investors value platforms largely on their reported growth, and governments use the same figures to size their digital economies and to tax online sellers. The trade is large: businesses in 43 economies sold almost US$27 trillion online in 2022, by the United Nations’ estimate (UNCTAD 2024).*

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BEFORE

> *Indonesia in 2023 offers a striking example, as its figures moved away from the commerce they stand for. Tokopedia, one of Indonesia's largest online marketplaces, saw its net revenue rise by 53 percent. Yet its transaction value, the value of what was bought through the platform, fell by 9 percent (GoTo Gojek Tokopedia 2024a, 2024b). Nationally, the statistics office, BPS, reported that online sales rose by about 41 percent (BPS 2025a), while the central bank's measure of online shopping fell by about 5 percent (Bank Indonesia 2023; KabarBursa 2024). Depending on which figure one reads, 2023 was either a boom or a stall.*

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BEFORE

> *There are three reasons why this thesis focuses on Indonesia. First, Indonesia allows the question to be tested. Several platforms report revenue and transaction value each year, and Badan Pusat Statistik (BPS), which is Indonesia's national statistics agency, asks online sellers when they began and how their sales changed. Second, the two sources already differ on scale: for 2024, BPS gives about US$81 billion for online sales and the central bank about US$31 billion (BPS 2025b; Bank Indonesia 2025). Third, new rules on platform commissions, discounts and tax act directly on these figures, so the evidence bears on how far those rules can reach.*

AFTER (~~deleted~~, added)

> *There are three reasons why this thesis focuses on Indonesia. First, Indonesia allows the question to be tested. Several platforms report revenue and transaction value each year, and ~~Badan Pusat Statistik (BPS), which is Indonesia's national statistics agency,~~* **BPS** *asks online sellers when they began and how their sales changed. Second, the two sources already differ on ~~scale.~~* **scale, probably because they cover different sellers:** *for*

2024, BPS gives about US$81 billion for online sales and the central bank about US$31 billion (BPS 2025b; Bank Indonesia 2025). Third, new rules on platform commissions, discounts and tax act directly on these figures, so the evidence bears on how far those rules can reach.

1.1 Research question and objectives

BEFORE

The analysis begins by measuring the invisible wedge. In 2023, three major platforms in Indonesia kept about US$3 billion of the US$43 billion that passed through them, so revenue shows only a small part of the commerce platforms carry, and marketplaces carry only about a fifth of online sales value in BPS's survey. The thesis then tests H1. In Indonesia, platform revenue grew faster than the commerce behind it, mainly because platforms kept a larger share of each sale, not because buyers bought more: Indonesian and regional platforms changed their take rates about three times as much as 27 platforms abroad. Next, the thesis turns to the national figures. National growth came mainly from more sellers, as H2 states. But the registered entry test fails, and about half of the 2023 rise in the count of sellers is not explained by first-time entry. Finally, the thesis finds that the new marketplace tax can collect from at most about 1 in 20 online sellers.

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1.2 Contributions

BEFORE

Third, for tax authorities and regulators, the thesis measures how far rules based on platform records reach. Reports from third parties improve tax compliance where they exist (Kleven et al. 2011; Pomeranz 2015), but such a rule reaches only the sellers who use platforms, so where most sellers work outside them, it covers a minority. The thesis also sets each recent rule, from the cap on ride-hailing commissions to the limit on discounts, against the figure it acts on. Several other countries have adopted similar rules (OECD 2020; European Union 2021), so the same question applies there.

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Chapter 2. Literature review

2.3 Digital records and third-party information

BEFORE

Tax research explains why such records matter. Reports from third parties improve compliance. In Denmark, evasion was close to zero on income that employers and banks reported, and substantial on income that taxpayers reported themselves (Kleven et al. 2011). In Chile, the paper trail of the value-added tax deters evasion, and enforcement spreads up the supply chain (Pomeranz 2015). In Brazil, rewards for asking for receipts raised the sales that firms reported (Naritomi 2019). Digital records can also hold information that conventional records lack (Berg, Burg, Gombovic and Puri 2020).

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2.4 Research gap

REMOVED

The shortfalls identified in Sections 2.1 to 2.3 define the research gap. Table 2.1 sets them side by side.

REMOVED

Table 2.1. What the literature establishes and where it stops

REMOVED

Section	*What it establishes*	*Main sources*	*Where it stops*
2.1 Platform economics and accounting	*Revenue is a fee that is a small share of the commerce, and the share can move with pricing, discounts and accounting*	*Rochet and Tirole 2003; Armstrong 2006; IASB 2014*	*Whether reported revenue stays a fair guide to reported transaction value has not been tested, and not for Indonesian platforms*

… (table of 3 rows)

BEFORE

> *Each shortfall becomes one question for the thesis, and Chapter 3 sets out the tools that answer each.*

AFTER (~~deleted~~, added)

> ~~*Each shortfall*~~ **The shortfalls identified in Sections 2.1 to 2.3 define the research gap, and each** *becomes one question for the* ~~*thesis, and*~~ **thesis.** *Chapter 3 sets out the tools that answer each.*

BEFORE

1. **How large is the difference between transaction value and revenue, and does it last?** *This follows from Section 2.1. It explains why the share is small and can move, but it does not test whether revenue stays a fair guide to the commerce.*
2. **Does the choice of figure change the conclusion about growth?** *This follows from Section 2.2. It shows that figures differ, but it does not test whether they grow for the same reasons, or what a rise in the count of sellers contains.*
3. **How far can rules built on platform records reach?** *This follows from Section 2.3. It shows that platform records are valuable, but not how much of online selling they cover.*

AFTER (~~deleted~~, added)

1. **How large is the difference between transaction value and revenue, and does it last?** *This follows from Section 2.1. It explains why the share is small and can move, but* ~~*it does not test*~~ *whether* **reported** *revenue stays a fair guide to* ~~*the commerce.*~~ **reported transaction value has not been tested, least of all for Indonesian platforms.**
2. **Does the choice of figure change the conclusion about growth?** *This follows from Section 2.2. It shows that figures differ, but* ~~*it does*~~ **platform and national figures have** *not* ~~*test whether they grow for the same reasons, or*~~ **been followed through time in one country, and published totals cannot show** *what a rise in the count of sellers contains.*
3. **How far can rules built on platform records reach?** *This follows from Section 2.3. It shows that platform records are valuable, but* ~~*not*~~ *how much of online selling they* ~~*cover.*~~ **cover has not been estimated for Indonesia.**

Chapter 3. Framework and hypotheses

Chapter 3. Framework and Hypotheses (opening)

BEFORE

1. **How large the difference between transaction value and revenue is, and whether it lasts:** *three measures, a split of revenue growth, and H1 (Sections 3.2 to 3.4).*
2. **Whether the choice of figure changes the conclusion about growth:** *H2, the entry test and a comparison of national figures (Sections 3.4 to 3.6).*
3. **How far rules built on platform records reach:** *how online selling divides between marketplaces and other channels (Section 3.6).*

AFTER (~~deleted~~, added)

1. **How large the difference between transaction value and revenue is, and whether it lasts:** *three measures,* ~~*a split*~~ **direct tests** *of* ~~*revenue growth,*~~ **what changes them,** *and H1 (Sections 3.2 to 3.4).*
2. **Whether the choice of figure changes the conclusion about growth:** *H2, the entry test and a comparison of national figures (Sections 3.4 to 3.6).*
3. **How far rules built on platform records reach:** *how online selling divides between marketplaces and other channels (Section 3.6).*

3.1 Terms and rules for comparison

NEW

> *Six terms recur throughout the thesis. Transaction value, V, is everything bought through a platform in a period, also called the commerce the platform carries. Companies call it gross merchandise value (GMV), gross transaction value (GTV) or total payment volume (TPV); the thesis treats these as one measure and notes where a company's scope differs. Revenue, R, is what the platform recognizes as its own income in its accounts. The take rate, m, is revenue divided by transaction value: the share of each sale that the platform keeps. Customer incentives are the discounts, vouchers and other promotions that a platform pays for. Accounting rules normally deduct them from revenue, and Tokopedia reports the deduction as a separate line.*

NEW

> *An online seller is a business that BPS counts as selling online; the text also calls it a seller or a business. The count of online sellers is BPS's estimate of how many there are in a year. First-time entry means that a seller began selling online in the year being measured. The measures built from these terms are defined in Section 3.2, and Appendix A lists every variable with its symbol, definition and source.*

REMOVED

> *Table 3.1 defines the terms used throughout the thesis, each once, with the symbol used in the formulas.*

REMOVED

Table 3.1. Terms and symbols

REMOVED

Term	*Symbol*	*Meaning*
Transaction value	*V*	*Everything bought through a platform in a period, also called the commerce the platform carries. Companies call it gross merchandise value (GMV), gross transaction value (GTV) or total payment volume (TPV); the thesis treats these as one measure and notes where a company's scope differs.*

… (table of 12 rows)

3.2 Three measures

BEFORE

where m_it = R_it / V_it is the take rate, the share of each sale that the platform keeps. The ecosystem ratio is the author's own measure, and it restates the take rate. An ecosystem ratio of 9 means that 9 units of transaction value lie outside revenue for every unit recorded as revenue. Equivalently, the platform keeps 10 percent of each sale.

AFTER (~~deleted~~, added)

where m_it = R_it / V_it is the take rate, the share of each sale that the platform keeps. The ecosystem ratio is the author's own measure, and it restates the take rate. An ecosystem ratio of 9 means that 9 units of transaction value lie outside revenue for every unit recorded as revenue. Equivalently, the platform keeps 10 percent of each sale. **Table 5.1 applies the measures to three platforms in Indonesia in 2023, where the wedge is about US$40 billion and the ecosystem ratio about 13.**

NEW

The take rate itself has two parts:

NEW

```
m_it = f_it − s_it                                   (5)
```

NEW

where f_it is the fee rate and s_it the incentive rate, both measured against transaction value. A take rate can rise because a platform charges more or because it spends less on discounts and vouchers. Accounting rules normally deduct such incentives from revenue (International Accounting Standards Board 2014). So a cut in incentives raises revenue even when no fee changes.

NEW

Equation (5) also shows where a rule acts. A cap on commissions acts on the fee rate, and a limit on discounts acts on the incentive rate. Chapter 6 sets Indonesia's recent rules against these parts of the take rate.

3.3 What changes the wedge and the ecosystem ratio (was: 3.3 Tracing a movement)

NEW

Equations (1) to (4) define the wedge and the ecosystem ratio. They do not say what moves them. Two direct tests do, and Chapter 5 reports both. The first tests whether revenue grows in proportion to transaction value, the claim of H1. The second asks what explains the size of the wedge and of the ecosystem ratio across platforms.

NEW

Revenue grows in proportion to transaction value when a 1 percent rise in transaction value comes with a 1 percent rise in revenue. That claim can be tested in growth rates, with g(·) as defined above:

NEW

$$g(R_{it}) = a + b\, g(V_{it}) + c\, Indonesia_i + d\, (Indonesia_i \times g(V_{it})) + u_{it} \qquad (6)$$

NEW

where Indonesia_i equals one for the Indonesian and regional platforms and zero for the foreign ones. If revenue is proportional to transaction value, b equals 1, a is close to zero, and the group does not matter, so c and d are zero. A slope below 1 in the Indonesian and regional group would mean that revenue moved with the platform's pricing and not with the commerce. The correlation between the two growth rates, which Chapter 5 reports for each group, gives the same test in one number.

NEW

The second test regresses the wedge and the ecosystem ratio on the features of a platform and of its home market:

NEW

$$Y_{it} = \alpha + \beta 1\, Indonesia_i + \beta 2\, \ln(1 + Listed_{it}) + \beta 3\, \ln(Assets_{it}) + \beta 4' Type_i + \gamma' Macro_{it} + \delta_t + \varepsilon_{it} \qquad (7)$$

NEW

where Y_it is the natural logarithm of the wedge in US dollars, or the ecosystem ratio. Listed_it is the number of years since the platform listed on a stock exchange. Assets_it is its total assets in US dollars, which measure size without using transaction value, because transaction value is part of the wedge itself. Type_i marks the kind of platform: marketplace, which is the base, on-demand, travel, freelance or resale. Macro_it holds four measures of the platform's home market: growth in gross domestic product (GDP), consumer-price inflation, growth in household spending and the central bank's policy rate. The year effects δ_t remove what all platforms share in a year.

NEW

The take rate and its parts do not appear on the right-hand side. The ecosystem ratio equals one divided by the take rate, minus one, so a regression of the ratio on the take rate would restate an identity and show nothing. Chapter 4 describes the data, the variables and the standard errors.

REMOVED

A large divergence calls for an explanation, and the same definitions supply it. Measured in logarithms, the growth of revenue splits exactly into the growth of transaction value and the change in the take rate:

REMOVED

$$\Delta \ln R_{it} = \Delta \ln V_{it} + \Delta \ln m_{it} \qquad (5)$$

REMOVED

where Δln X is the change in the natural logarithm of X from one year to the next. The first term is the volume effect, more or less commerce. The second is the take-rate effect, a larger or smaller share of each sale. A log change treats a rise and a fall by the same factor as equal in size, so the two effects can be compared directly. The take rate itself has two parts:

REMOVED

$$m_{it} = f_{it} - s_{it} \qquad (6)$$

REMOVED

where f_it is the fee rate and s_it the incentive rate, both measured against transaction value. A take rate can rise because a platform charges more or because it spends less on discounts and vouchers. Accounting rules normally deduct such incentives from revenue (International Accounting Standards Board 2014). So a cut in incentives raises revenue even when no fee changes.

REMOVED

Equations (5) and (6) together divide a movement in revenue into more commerce, higher fees and lower incentives. A change in the mix of a company's businesses is a further source, read from its reports where they describe one. The split shows how a movement happened. It does not show why the company chose it, and company explanations are read as reports.

REMOVED

Equation (6) also shows where a rule acts. A cap on commissions acts on the fee rate, and a limit on discounts acts on the incentive rate. Chapter 6 sets Indonesia's recent rules against these parts of the take rate.

3.4 Hypotheses

BEFORE

Two hypotheses follow from the research question. H1 asks whether revenue follows the commerce carried within a platform. H2 asks where national online growth comes from, which decides whether the commerce on platforms can stand for national online commerce. Together they test the two links in Figure 3.1. Table 3.2 pairs each hypothesis, and the entry test of Section 3.5, with its main test and with the result that would count against it.

AFTER (~~deleted~~, added)

Two hypotheses follow from the research question. H1 asks whether revenue follows the commerce carried within a platform. H2 asks where national online growth comes from, which decides whether the commerce on platforms can stand for national online commerce. Together they test the two links in Figure 3.1. ~~Table 3.2 pairs each~~ **Each** *hypothesis, and the entry test of Section 3.5,* **comes** *with its main test and with the result that would count against it.*

REMOVED

Table 3.2. Hypotheses, tests and what would count against them

REMOVED

Test	*What it asks*	*Main test*	*What would count against it*
H1	*Do revenue and transaction value grow together within a platform?*	*Size of yearly changes in the take rate, Indonesian and regional platforms against listed platforms abroad*	*Changes in Indonesia far larger than abroad*

… (table of 3 rows)

BEFORE

Growth rates never match exactly, so H1 is tested by comparison. The test compares the size of yearly changes in the take rate, measured in logs as in equation (5), between two groups. One group is the Indonesian and regional platforms. The other is listed platforms abroad, where revenue and transaction value broadly move together. Changes far larger in Indonesia would count against H1. Chapter 4 describes the two groups and the test.

AFTER (~~deleted~~, added)

~~Growth rates never match exactly, so~~ H1 is tested ~~by comparison.~~ **in two ways.** *The ~~test~~* **first is equation (6), an ordinary least squares (OLS) regression of revenue growth on transaction value growth, and H1 holds in a group if the slope is 1. The second** *compares the size of yearly changes in the take rate, measured ~~in logs~~ as ~~in equation (5),~~* **simple percentage changes,** *between two groups. One group is the Indonesian and regional platforms. The other is listed platforms abroad, where revenue and transaction value broadly move together. ~~Changes~~* **A slope far below 1 in Indonesia, or changes** *far larger in Indonesia* **than abroad,** *would count against H1. Chapter 4 describes the two groups and the ~~test.~~* **tests.**

BEFORE

> *BPS publishes both the number of businesses selling online, N, and their total online sales value, Y. Since Y equals N times the sales per business, its growth in logarithms splits exactly in two:*

AFTER (~~deleted~~, added)

> *BPS publishes both the number of businesses selling online, N, and their total online sales value, Y. Since Y equals N times the sales per business, its growth ~~in logarithms~~ splits ~~exactly~~ in two:*

NEW

```
1 + g(Y_t) = (1 + g(N_t)) × (1 + g(Y_t / N_t))                    (8)
```

REMOVED

```
Δln Y_t = Δln N_t + Δln (Y_t / N_t)                    (7)
```

BEFORE

> *where the first term is the extensive margin and the second the intensive margin. H2 holds if the first term is more than half of the total. The second hypothesis states the case in which more businesses drive growth.*

AFTER (~~deleted~~, added)

> *where ~~the first term~~* **g(N)** *is the* **growth in the number of businesses, the** *extensive ~~margin~~* **margin,** *and* **g(Y/N)** *the ~~second~~* **growth in sales per business,** *the intensive margin. H2 holds if* **g(N) is larger than g(Y/N), which is** *the ~~first term is~~* **same as saying that** *more ~~than half~~* **businesses account for most** *of the ~~total.~~* **growth.** *The second hypothesis states the case in which more businesses drive growth.*

3.6 Further analyses

BEFORE

> *Four further analyses complete the framework without forming hypotheses of their own. Each answers a part of the research question that H1, H2 and the entry test leave open. Table 3.3 lists them.*

AFTER (~~deleted~~, added)

> *Four further analyses complete the framework without forming hypotheses of their own. Each answers a part of the research question that H1, H2 and the entry test leave open. ~~Table 3.3 lists them.~~* **They are the tracing of large movements, the market values of listed platforms, the channels and reach of platform records, and the comparison of national figures.**

REMOVED

> **Table 3.3. Further analyses**

REMOVED

Analysis	*What it asks*	*Main evidence*
Tracing large movements	*What moved revenue when it parted from transaction value?*	*Equations (5) and (6), applied to company reports*

… (table of 4 rows)

BEFORE

> *Tracing large movements applies the split of Section 3.3. Wherever H1 finds a large change in the take rate, the thesis divides the change in revenue into its parts. It also reads the company's reports for changes in the mix of businesses and in reporting scope.*

AFTER (~~deleted~~, added)

> *Tracing large movements ~~applies the split of Section 3.3.~~* **asks what moved revenue when it parted from transaction value.** *Wherever H1 finds a large change in the take rate, the thesis divides the change in revenue into ~~its parts.~~* **more commerce, higher fees and lower incentives, using company reports (Appendix B).** *It also reads the ~~company's~~ reports for changes in the mix of businesses and in reporting scope.*

BEFORE

> *The market-value analysis asks whether investors, who price revenue (Section 2.1), treat its two sources alike. Equation (5) shows that revenue can grow in two ways, through more commerce or through a larger take rate.*

AFTER (~~deleted~~, added)

> *The market-value analysis asks whether investors, who price revenue (Section 2.1), treat its two sources alike. Equation ~~(5)~~* **(4)** *shows that revenue ~~can grow~~* **growth has two sources: growth** *in ~~two ways, through more commerce or through~~* **transaction value and** *a ~~larger~~* **change in the** *take rate.*

BEFORE

```
Δln MV_it = β_V Δln V_it + β_m Δln m_it + year effect + error      (8)
```

AFTER (~~deleted~~, added)

```
Δln MV_it = β_V Δln V_it + β_m Δln m_it + year effect + error ~~(8)~~ **(9)**
```

BEFORE

> *where β_V shows how market value moves with transaction value, and β_m how it moves with the take rate. Two predictions were written down before the market values were collected. The first is that market value moves with transaction value, so β_V is positive. The second is that markets value growth from more commerce more than growth from a higher take rate, so β_V is larger than β_m.*

AFTER (~~deleted~~, added)

where β_V shows how market value moves with transaction value, and β_m how it moves with the take rate. Two predictions were written down before the market values were collected. The first is that market value moves with transaction value, so β_V is positive. The second is that markets value growth from more commerce more than growth from a higher take rate, so β_V is larger than β_m. **The predictions were registered with changes measured in logs, and the specification is kept as registered.**

Chapter 4. Data and methods

4.1 The platform data

BEFORE

Two groups of platforms are compared. The Indonesian and regional group has five platforms, listed in Table 4.1. Three are Indonesian marketplaces, and two are the on-demand businesses of Grab and GoTo, which cover rides and deliveries. The foreign group has 27 listed platforms in marketplaces, travel and ticketing, freelance work, resale and food delivery. Eight were chosen first, and 19 were added to widen the comparison. Ten more were examined and left out, six because they sell mostly their own goods and four because they do not disclose transaction value.

AFTER (~~deleted~~, added)

Two groups of platforms are compared. The Indonesian and regional group has five platforms, listed in ~~Table 4.1.~~ **Appendix B.** *Three are Indonesian marketplaces, and two are the on-demand businesses of Grab and GoTo, which cover rides and deliveries. The foreign group has 27 listed platforms in marketplaces, travel and ticketing, freelance work, resale and food delivery. Eight were chosen first, and 19 were added to widen the comparison. Ten more were examined and left out, six because they sell mostly their own goods and four because they do not disclose transaction value.*

REMOVED

Table 4.1. The Indonesian and regional group of platforms

REMOVED

Platform series	*What it covers*	*Years*	*Yearly pairs*	*How it is labelled*
Tokopedia	*The e-commerce segment of GoTo, an Indonesian marketplace*	*2022–2023*	*1*	*Direct segment disclosure*

… (table of 5 rows)

BEFORE

Table 4.1 states the scope of each series because the series differ, and only Tokopedia is an Indonesian segment alone. The thesis never adds the series together, and each platform enters the tests on its own. Sea, whose Shopee business has Indonesia as a main market, sits in the foreign group.

AFTER (~~deleted~~, added)

> ~~*Table 4.1*~~ **Appendix B** *states the scope of each series because the series differ, and only Tokopedia is an Indonesian segment alone. The thesis never adds the series together, and each platform enters the tests on its own. Sea, whose Shopee business has Indonesia as a main market, sits in the foreign group.*

BEFORE

> *Two platforms sit close to that line. Zalando is one of the eight and sells part of what it carries, so Appendix B repeats the test without it. One added pair, Ozon from 2021 to 2022, ends in a kept year but starts in a dropped one. Appendix B repeats the test without Ozon as well. The result is 15 pairs for the Indonesian and regional group and 158 for the foreign group.*

AFTER (~~deleted~~, added)

> *Two platforms sit close to that line. Zalando is one of the eight and sells part of what it carries, so Appendix ~~B~~* **C** *repeats the test without it. One added pair, Ozon from 2021 to 2022, ends in a kept year but starts in a dropped one. Appendix ~~B~~* **C** *repeats the test without Ozon as well. The result is 15 pairs for the Indonesian and regional group and 158 for the foreign group.*

4.3 Testing H1

NEW

> *The test of H1 has two parts, run on the same 32 platforms: the five Indonesian and regional platforms, Tokopedia, Blibli, Bukalapak and the on-demand businesses of Grab and GoTo, and the 27 platforms abroad.*

NEW

> *The first part is equation (6). Each observation is a pair of consecutive years of one platform, and a missing year is never filled. Growth is the simple percentage change, and growth rates are capped at their 5th and 95th percentiles (Section 4.4). The regression runs on all platforms with the group interaction, on the foreign group alone and on the Indonesian and regional group alone, with standard errors clustered by platform. The thesis also reports the correlation between revenue growth and transaction value growth for each group. Appendix C gives the rank correlation and the correlation without caps.*

REMOVED

> *The test of H1 asks how far each platform's take rate moves from one year to the next. For each pair of years it takes the change in the logarithm of the take rate, the take-rate term of equation (5). This change is growth divergence measured in logs, and it does not depend on how fast transaction value grew. Tokopedia's log change in 2023 is 0.52, which is a rise by a factor of 1.68, or 68 percent.*

BEFORE

A move in either direction counts against H1, so the test uses the size of each yearly change in the take rate. Each platform gets one number, the median size of its yearly changes, because the pairs of one platform are not independent of one another.

AFTER (~~deleted~~, added)

The second part asks how far each platform's take rate moves from one year to the next. For each pair of years it takes the simple percentage change in the take rate. Tokopedia's change in 2023 is 68 percent. *A move in either direction counts against H1, so the test uses the size of each yearly ~~change in the take rate.~~* **change.** *Each platform gets one number, the median size of its yearly changes, because the pairs of one platform are not independent of one another.*

BEFORE

Appendix B lists the checks on this test. Some change the sample. They drop each platform in turn, drop two at a time, treat Tokopedia and GoTo as one company, move Sea to the regional group, keep only the first five years after listing, or narrow the test to the three marketplaces. Others change the measure: growth divergence in points as Chapter 3 defines it, exact tests, the change in the take rate in percentage points (to see whether a thin take rate explains the result), and the correlation of revenue growth with transaction value growth. One repeats the test with quarterly pairs.

AFTER (~~deleted~~, added)

Appendix ~~B~~ **C** *lists the checks on this test. Some change the sample. They drop each platform in turn, drop two at a time, treat Tokopedia and GoTo as one company, move Sea to the regional group, keep only the first five years after listing, or narrow the test to the three marketplaces. Others change the measure:* **the change in logs, which the first version of this test used,** *growth divergence in points as Chapter 3 defines it, exact tests,* **and** *the change in the take rate in percentage ~~points (to~~* **points, to** *see whether a thin take rate explains the ~~result), and the correlation of revenue growth with transaction value growth.~~* **result.** *One repeats the test with quarterly pairs.*

4.4 What explains the wedge and the ecosystem ratio (was: 4.4 Tracing the moves)

NEW

Equation (7) is estimated on a panel of 213 platform-years for the same 32 platforms, 20 of them Indonesian and regional platform-years and 193 foreign. A platform-year enters when the company reports both transaction value and revenue for it under the rules of Section 3.1, and a year that fails them stays missing. Amounts are in US dollars at the World Bank's period-average rate for the year (World Bank 2026).

NEW

Size is the platform's total assets, taken from annual reports filed with the US Securities and Exchange Commission for 26 platforms (U.S. Securities and Exchange Commission 2026) and from published financial statements for the other six: Tokopedia and GoTo, which share one group statement, Blibli, Bukalapak, Rakuten and Zalando. Total assets are those of the company as a whole. They overstate the platform business at GoTo,

Rakuten and Sea. The number of years since listing counts from the year the platform listed on a stock exchange, from public record, and a year before listing counts as zero. The type of platform is the author's classification (Appendix B).

NEW

The home market is the country of the platform's main market. It is the United States for most US-listed platforms and Indonesia for the five Indonesian and regional platforms. It is also Indonesia for Sea, whose Shopee business has Indonesia as a main market. GDP growth, inflation and growth in household spending come from the World Bank (2026). The policy rate is the central bank's rate from the Bank for International Settlements (2026), averaged over the year. It is not available for Kazakhstan or Nigeria, so the regressions with macroeconomic variables have 178 platform-years from 30 platforms.

NEW

Growth rates and the ecosystem ratio are capped at their 5th and 95th percentiles across all platforms, the rule that Section 4.5 uses for market value. Standard errors are clustered by platform, with the small-sample correction, and p-values use a t distribution with one fewer degrees of freedom than platforms. Only five platforms are Indonesian and regional, so the thesis adds a wild cluster bootstrap p-value for the Indonesia coefficient (Cameron, Gelbach and Miller 2008). Appendix C lists the checks on this coefficient. All of these tests are exploratory, because they were designed after the data were seen.

REMOVED

The second objective, to reconcile large movements in revenue with disclosed changes in fees, discounts and scope, is answered in two steps. The first divides each movement into volume and take rate with equation (5). The take-rate share is the size of the take-rate change divided by the sum of the sizes of both changes. The same split is made for the wedge.

BEFORE

The second step splits the take-rate change with equation (6). Some companies disclose both their gross fees and their incentives. For them, the thesis splits the growth of revenue over a window into three parts: more commerce, a change in the fee rate and a change in the incentive rate. A change in the mix of businesses is recorded from the company's reports.

AFTER (~~deleted~~, added)

The second ~~step splits the take-rate change~~ **objective, to reconcile large movements in revenue** *with ~~equation (6). Some companies disclose~~* **disclosed changes in fees, discounts and scope, rests on company reports. Where a company discloses** *both ~~their~~* **its** *gross fees and ~~their incentives. For them,~~* **its incentives,** *the thesis splits the growth of revenue over a window into ~~three parts:~~ more commerce, a change in the fee*

rate and a change in the incentive ~~rate.~~ **rate (equation (5)). The split is exact only in logs, so Appendix B keeps the log definition for these windows.** *A change in the mix of businesses is recorded from the company's reports.*

REMOVED

The sources differ by company. Tokopedia's audited segment note gives gross revenue, incentives and net revenue, and the thesis checks them against the audited consolidated statements. For Tokopedia the thesis also reports the change line by line in rupiah, the usual way to read accounts, and says where the two views differ.

BEFORE

Grab is read from its annual filings and GoTo from its audited statements. Blibli does not report incentives, so its split rests on a proxy built from its releases. Seven windows are traced, and Appendix A lists them. The thesis also records where accounting treatment or reporting scope matters, such as the deduction of incentives from revenue and the travel business inside Blibli's retail segment. Company explanations are read as reports and are not tested.

AFTER (~~deleted~~, added)

The sources differ by company. Tokopedia's audited segment note gives gross revenue, incentives and net revenue, and the thesis checks them against the audited consolidated statements. For Tokopedia the thesis also reports the change line by line in rupiah, the usual way to read accounts, and says where the two views differ. *Grab is read from its annual filings and GoTo from its audited statements. Blibli does not report incentives, so its split rests on a proxy built from its releases. Seven windows are traced, and Appendix ~~A~~* **B** *lists them. The thesis also records where accounting treatment or reporting scope matters, such as the deduction of incentives from revenue and the travel business inside Blibli's retail segment. Company explanations are read as reports and are not tested.*

4.5 Market values

BEFORE

The market-value analysis of Section 3.6 uses year-end market values from the financial data service LSEG, under the university licence. The main sample has 28 listed platforms and 138 platform-years, made up of Bukalapak and 27 platforms abroad, and it uses whole-company figures only. It is not the list of 27 used for H1. It adds Airbnb, Booking, DoorDash, Lyft and Uber, and it lacks Despegar, Mercado Libre, Ozon, Rakuten and Sea.

AFTER (~~deleted~~, added)

The market-value analysis of Section 3.6 uses year-end market values from the ~~financial~~ data service ~~LSEG,~~ **of the London Stock Exchange Group (LSEG),** *under the university licence. The main sample has 28 listed platforms and 138 platform-years, made up of Bukalapak and 27 platforms abroad, and it uses whole-company figures only. It is not the list of 27 used for H1. It adds Airbnb, Booking, DoorDash, Lyft and Uber, and it lacks Despegar, Mercado Libre, Ozon, Rakuten and Sea.*

BEFORE

> *Equation (8) is estimated with a fixed effect for each year, which removes what all platforms share in that year. Standard errors are clustered by company, which allows for the repeated observations of one company. Each prediction names a direction, so both tests are one-sided. The test was written down before the market values were pulled. The result is an association, and market value also changes when a company issues shares, so the thesis does not call it a stock return. A second run adds five segment series and Mercado Libre as a robustness check. Grab, GoTo and Sea enter only there, because their reported transaction value covers only part of the business that the market value prices.*

AFTER (~~deleted~~, added)

> *Equation ~~(8)~~* **(9)** *is estimated with a fixed effect for each year, which removes what all platforms share in that year. Standard errors are clustered by company, which allows for the repeated observations of one company. Each prediction names a direction, so both tests are one-sided. The test was written down before the market values were ~~pulled.~~* **pulled, with changes measured in logs, and it keeps that definition.** *The result is an association, and market value also changes when a company issues shares, so the thesis does not call it a stock return. A second run adds five segment series and Mercado Libre as a robustness check. Grab, GoTo and Sea enter only there, because their reported transaction value covers only part of the business that the market value prices.*

4.6 The survey files and H2

BEFORE

> *H2 uses the published figures and the split of equation (7). BPS prints two counts of businesses for 2023, so the thesis computes the 2023 to 2024 split with both and reports both. The 2022 to 2023 split uses the count that the survey weights match.*

AFTER (~~deleted~~, added)

> *H2 uses the published figures and the split of equation ~~(7).~~* **(8).** *BPS prints two counts of businesses for 2023, so the thesis computes the 2023 to 2024 split with both and reports both. The 2022 to 2023 split uses the count that the survey weights match.*

BEFORE

> *The published figures cannot tell a new seller from one counted for the first time, but the survey files can. Table 4.2 lists them. Each row is a business that sold online in the activity year. Its survey weight shows how many businesses it represents, so the weights add up to the national count.*

AFTER (~~deleted~~, added)

> *The published figures cannot tell a new seller from one counted for the first time, but the survey files can. ~~Table 4.2 lists them.~~* **The three files are the E-Commerce rounds of 2021, 2023 and 2024, which describe the activity years 2020, 2022 and 2023.**[1] *Each row is a business that sold online in the activity year. Its survey weight shows how many businesses it represents, so the weights add up to the national count.*

NEW

REMOVED

Table 4.2. The three BPS survey files

REMOVED

Survey round	*Activity year*	*Rows*	*Weighted total*	*BPS count*
E-Commerce 2021	*2020*	*11,848*	*2,361,423*	*2,361,423*

… (table of 3 rows)

REMOVED

**Implied by the published count for 2023 and its published growth of 27.4 percent.*

4.9 Comparing national figures

BEFORE

The third objective, to compare the platform evidence with national and payment figures, rests on the sources that Table 4.3 lists, with what each one counts. Beyond the rules of Section 3.1, the comparison is closest where the coverage of two sources overlaps. Payment figures serve as context, because they show how people pay and not what they buy online.

AFTER (~~deleted~~, added)

The third objective, to compare the platform evidence with national and payment figures, rests on the sources that ~~Table 4.3~~ **Appendix A** *lists, with what each one counts. Beyond the rules of Section 3.1, the comparison is closest where the coverage of two sources overlaps. Payment figures serve as context, because they show how people pay and not what they buy online.*

REMOVED

Table 4.3. Sources compared in Objective 3

REMOVED

Source	*What it covers*	*How it counts*
Platform reports	*Everything bought through one platform*	*The platform's own records*

… (table of 7 rows)

4.10 Labels, traceability and licensed data

BEFORE

The thesis also separates tests by when they were written. A registered test was written down, dated and stored in the repository before the data were opened. This applies to the entry test, the scenario rules for value, the comparison with Bank Indonesia and the market-value test, among others. Appendix C lists them with their dates. An exploratory test was designed after some results were seen. The test with the larger groups in Section 4.3 belongs here, because it was set up after a first test on the three marketplaces. A descriptive result reports numbers without a test.

AFTER (~~deleted~~, added)

The thesis also separates tests by when they were written. A registered test was written down, dated and stored in the repository before the data were opened. This applies to the entry test, the scenario rules for value, the comparison with Bank Indonesia and the market-value test, among others. Appendix ~~C~~ **D** *lists them with their dates. An exploratory test was designed after some results were seen. The test with the larger groups in Section 4.3 belongs here, because it was set up after a first test on the three marketplaces. A descriptive result reports numbers without a test.*

BEFORE

Three sources are licensed and are not in the repository: the BPS survey files, the LSEG market values and the World Bank survey files. The repository records the size and checksum of each file, so anyone who obtains the same files can check them. It holds the code and the national totals it produces, and no business-level record. BPS is cited as the source of every survey figure, in line with the data-use agreement. Chapter 7 states what these data cannot show.

AFTER (~~deleted~~, added)

The World Bank indicators, the Bank for International Settlements policy rates and the total assets from the filings with the US Securities and Exchange Commission are public. The repository holds the downloaded values and the scripts that fetched them. *Three sources are licensed and are not in the repository: the BPS survey files, the LSEG market values and the World Bank survey files. The repository records the size and checksum of each file, so anyone who obtains the same files can check them. It holds the code and the national totals it produces, and no business-level record. BPS is cited as the source of every survey figure, in line with the data-use agreement. Chapter 7 states what these data cannot show.*

Chapter 5. Results

Chapter 5. Results (opening)

BEFORE

Every result in this chapter carries one of the three labels of Section 4.10: registered, exploratory or descriptive. A result given without a label is descriptive. Table 5.1 gives each main result with its label, the checks behind it and the section that reports it.

AFTER (~~deleted~~, added)

Every result in this chapter carries one of the three labels of Section 4.10: registered, exploratory or descriptive. A result given without a label is descriptive. ~~Table 5.1 gives each main~~ **Each section states the label next to the** *result ~~with its label, the checks behind~~ it ~~and the section that reports it.~~* **reports.**

REMOVED

Table 5.1. Main results and the evidence behind them

REMOVED

Result	*Label*	*Main checks*	*Section*
Inner link: revenue and platform commerce			

… (table of 16 rows)

5.1 How much revenue leaves out

BEFORE

Revenue showed only a small part of what Indonesia's major platforms carried. In 2023, Tokopedia, Shopee and Grab together carried about US$43 billion of transaction value and kept about US$3 billion as revenue. Revenue was about 7 percent of that commerce, and the invisible wedge was about US$40 billion. The ecosystem ratio was about 13, so almost 13 dollars of commerce passed outside revenue for every dollar the platforms kept. Table 5.2 shows each platform.

AFTER (~~deleted~~, added)

Revenue showed only a small part of what Indonesia's major platforms carried. In 2023, Tokopedia, Shopee and Grab together carried about US$43 billion of transaction value and kept about US$3 billion as revenue. Revenue was about 7 percent of that commerce, and the invisible wedge was about US$40 billion. The ecosystem ratio was about 13, so almost 13 dollars of commerce passed outside revenue for every dollar the platforms kept. Table ~~5.2~~ **5.1** *shows each platform.*

BEFORE

Table 5.2. The invisible wedge of three platforms, 2023 (US$ billion)

AFTER (~~deleted~~, added)

Table ~~5.2.~~ *5.1.** The invisible wedge of three platforms, 2023 (US$ billion)***

NEW

This table reports transaction value, revenue, the wedge (transaction value less revenue) and the ecosystem ratio (the wedge divided by revenue) for Tokopedia, Shopee in Indonesia and Grab in Indonesia in 2023. The last column states how each platform's amounts are obtained. The total row is the sum of the three platforms and is not a national total.

BEFORE

The size of the wedge does not depend on the assumed rates. A take rate for Shopee between 9 and 11 percent keeps the total near US$40 billion, and dropping any one platform still leaves more than US$20 billion. For platforms that report both amounts, the wedge was 97 to 99 percent of transaction value in every year. It moved almost one for one with the commerce, and Appendix A gives the wedge by platform and year.

AFTER (~~deleted~~, added)

*The size of the wedge does not depend on the assumed rates. A take rate for Shopee between 9 and 11 percent keeps the total near US$40 billion, and dropping any one platform still leaves more than US$20 billion. For platforms that report both amounts, the wedge was 97 to 99 percent of transaction value in every year. It moved almost one for one with the commerce, and Appendix ~~A~~ **B** gives the wedge by platform and year.*

5.2 Testing H1

NEW

Revenue followed transaction value abroad and did not in Indonesia and the region, so the thesis rejects H1 for the Indonesian and regional platforms. Table 5.2 describes the panel behind the test. The typical Indonesian and regional platform-year has an ecosystem ratio of about 35, against about 5.5 abroad. Transaction value grew by a median of 16 percent a year in both groups, and revenue by 38 percent in Indonesia and the region against 21 percent abroad. The Indonesian and regional platforms are also younger, a median of one year since listing against four.

NEW

Table 5.2. Summary statistics and correlations

NEW

This table reports summary statistics and pairwise correlations for the panel of 213 platform-years used in Sections 5.2 and 5.3, which covers 32 platforms. Panel A gives the number of observations (N), the mean, the median (P50) and the standard deviation (SD) of each variable for the Indonesian and regional platforms and for the foreign platforms. Panel B gives pairwise Pearson correlations for all platforms. Growth rates and the ecosystem ratio are capped at their 5th and 95th percentiles; the wedge and total assets are in US$ million. Appendix A defines the variables. ∗∗∗, ∗∗ and ∗ correspond to statistical significance at the 1%, 5% and 10% levels.

NEW

Panel A. Summary statistics

NEW

Variable	*Indonesian and regional*				*Foreign*			
	N	**Mean**	**P50**	**SD**	**N**	**Mean**	**P50**	**SD**

… (table of 11 rows)

NEW

Panel B. Pairwise correlations

NEW

Variables	*(1)*	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*	*(8)*	*(9)*
(1) ln(Wedge)	*1*								

… (table of 9 rows)

NEW

Note: Inflation and household spending growth are left out of Panel B to fit the page; they correlate 0.80 with the policy rate and 0.92 with GDP growth. The policy rate is missing for Kazakhstan and Nigeria, and some growth and macroeconomic values are missing for the first year of a platform or for 2025. Source: company filings; World Bank; Bank for International Settlements; author's calculations. Exploratory.

NEW

Across all platforms, revenue growth and transaction value growth have a correlation of 0.84 (Panel B). The wedge and transaction value correlate at 0.9994 in levels, as they must, because the wedge is transaction value minus revenue and revenue is a small part of it. So the test of H1 uses growth rates.

NEW

Table 5.3 tests H1 in growth rates. Abroad, a rise of 1 percentage point in the growth of transaction value goes with a rise of 0.99 points in the growth of revenue (standard error 0.07), which cannot be told from the 1 that proportional growth requires (p = 0.84). The two growth rates correlate at 0.90. In Indonesia and the region the slope is 0.40 (standard error 0.16) and differs from 1 (p = 0.02). The correlation is 0.31, and the constant is 0.44: revenue grew by about 44 percent a year whatever transaction value did. In the regression on all platforms, the slope of the Indonesian and regional group is 0.59 below the foreign slope. The Indonesian and regional result rests on 15 pairs from five platforms, so its standard errors are less reliable than those abroad.

NEW

Table 5.3. Revenue growth and transaction value growth

NEW

This table reports OLS estimates of equation (6). The dependent variable is the growth of revenue, g(R), the simple percentage change from the previous year. Column (1) uses all platforms, with an interaction between transaction value growth and the Indonesia and region dummy, so the coefficient on g(V) in that column is the slope of the foreign platforms. Column (2) uses the 26 foreign platforms with a pair of consecutive years, and

column (3) the five Indonesian and regional platforms. Growth rates are capped at their 5th and 95th percentiles. Appendix A defines the variables. Standard errors (in parentheses) are clustered at the platform level. ∗∗∗, ∗∗ and ∗ correspond to statistical significance at the 1%, 5% and 10% levels.

NEW

Dependent variable	*g(R)*	*g(R)*	*g(R)*
Sample	*All platforms*	*Foreign*	*Indonesian and regional*

… (table of 15 rows)

NEW

Note: The slope on g(V) is 1 if revenue grows in proportion to transaction value. Source: company filings; author's calculations. Exploratory.

REMOVED

Revenue did not keep pace with transaction value at the Indonesian and regional platforms, and the thesis rejects H1. At the typical platform in this group, the take rate moved by about 20 percent a year, up or down. At the typical foreign platform it moved by about 6 percent, so the moves were about three times as large. Table 5.3 gives the test and its checks.

REMOVED

Table 5.3. The test of H1

REMOVED

Test	*Method*	*Indonesian and regional group*	*Foreign group*	*p-value*
Yearly change in log take rate, 5 platforms against 27	*Mann–Whitney*	*0.20*	*0.06*	*0.004*

… (table of 5 rows)

REMOVED

Note: Group values are medians of each platform's median yearly change. Appendix B lists every check, including the tests without Ozon and Zalando.

BEFORE

The main test compares the five platforms with 27 abroad, using one median per platform. Both methods of Section 4.3 reject equal moves (p = 0.004 and 0.022), and exact versions that list every possible split of the 32 platforms agree (p = 0.004 and 0.021). No single platform carries the result: dropping any one, or treating Tokopedia and GoTo as one company, keeps p at 0.016 or below. Dropping the two largest movers, Blibli and Tokopedia, weakens it to p = 0.058 (Appendix B).

AFTER (~~deleted~~, added)

> *The ~~main~~* **earlier** *test ~~compares~~* **of** *the ~~five platforms with 27 abroad, using one median per platform. Both methods~~* **size** *of ~~Section 4.3 reject equal~~* **take-rate** *moves* **points the same way. At the typical Indonesian and regional platform the take rate moved by 22 percent a year, up or down. At the typical foreign platform it moved by 6 percent, so the moves were about 3.6 times as large (Mann–Whitney p = 0.003; shuffling the group labels, p = 0.01). Measured in logs, as in an earlier version of this test, the medians are 0.20 and 0.06** *(p = ~~0.004 and 0.022),~~* **0.004),** *and exact versions that list every possible split of the 32 platforms agree (p = 0.004 and 0.021). No single platform carries ~~the~~* **that** *result: dropping any one, or treating Tokopedia and GoTo as one company, keeps p at 0.016 or below. Dropping the two largest movers, Blibli and Tokopedia, weakens it to p = 0.058 (Appendix ~~B).~~* **C).**

BEFORE

> *The main test was set up after a first, narrower test, so it is exploratory. The narrower test, set before the wider one, compares the three Indonesian marketplaces with the eight foreign platforms chosen first. It points the same way. With one median per platform, the Mann–Whitney test rejects equal moves (p = 0.024). A shuffle of platform labels comes close without passing (p = 0.06), since three platforms give a test little power. In that sample, revenue growth and transaction value growth had a correlation of 0.10 in Indonesia and 0.98 abroad, excluding Sea.*

AFTER (~~deleted~~, added)

> *The ~~main~~ test* **of the size of moves** *was set up after a first, narrower test, so it is exploratory. The narrower test, set before the wider one, compares the three Indonesian marketplaces with the eight foreign platforms chosen first. It points the same way. With one median per platform, the Mann–Whitney test rejects equal moves (p = 0.024). A shuffle of platform labels comes close without passing (p = 0.06), since three platforms give a test little power. In that sample, revenue growth and transaction value growth had a correlation of 0.10 in Indonesia and 0.98 abroad, excluding Sea.*

BEFORE

> *The main test and the narrower test ask about the size of the moves, not their direction, and direction is less telling. Revenue grew faster than transaction value in 13 of the 15 Indonesian and regional pairs, and in two of them the two moved in opposite directions. Abroad, take rates also rose in about two thirds of pairs, but by much less. What sets the Indonesian and regional platforms apart is the size of the moves.*

AFTER (~~deleted~~, added)

> ~~*The main test and the narrower test ask about*~~ **Both tests of** *the size of ~~the moves,~~* **moves ask about their size,** *not their direction, and direction is less telling. Revenue grew faster than transaction value in 13 of the 15 Indonesian and regional pairs, and in two of them the two moved in opposite directions. Abroad, take rates also rose in about two thirds of pairs, but by much less. What sets the Indonesian and regional platforms apart is the size of the moves.*

BEFORE

> *The large moves were not simply a mark of young platforms, an exploratory check. The Indonesian and regional platforms made their largest moves one or two years after listing on the stock market. Foreign platforms, by contrast, did not move their take rates more in their first three years after listing than later (median 0.07 against 0.08, p = 0.24). This comparison uses the 15 foreign platforms with years in both periods. Nor did foreign platforms in lower-income markets move more in 2022 to 2024: one of seven did, against all four Indonesian and regional platforms with enough years. The repricing was an Indonesian and regional episode of those years.*

AFTER (~~deleted~~, added)

> ~~*The large moves were*~~ **Age does** *not* ~~*simply a mark of young platforms, an exploratory check.*~~ **explain the result.** *The Indonesian and regional platforms made their largest moves one or two years after listing on the stock* ~~*market. Foreign platforms, by contrast,*~~ **market, but foreign platforms** *did not move their take rates more in their first three years after listing than later (median 0.07 against* ~~*0.08,*~~ **0.08 in logs,** *p =* ~~*0.24). This comparison uses*~~ **0.24, an exploratory check), and Table 5.4 shows that years since listing explains neither** *the* ~~*15 foreign platforms with years in both periods. Nor did foreign platforms in lower-income markets move more in 2022 to 2024: one of seven did, against all four Indonesian and regional platforms with enough years. The repricing was an Indonesian and regional episode of those years.*~~ **wedge nor the ecosystem ratio.**

REMOVED

> *The thin take rate of Indonesian and regional platforms explains much of the threefold difference, an exploratory finding. Measured in percentage points, both groups changed their take rates by about 1 point a year, and the difference between them is nowhere near large enough to count (p = 0.65).*

BEFORE

> *The two groups differ in the take rate they start from. The typical Indonesian and regional platform keeps about 1.6 percent of each sale, against about 17 percent abroad. The thinnest take rates belong to the three marketplaces, while Grab and GoTo keep shares closer to those abroad. As Section 3.2 showed, the same 1-point rise then has very different effects. It lifts revenue by more than half at the thinner take rate and by about 6 percent at the thicker one. Figure 5.1 shows the three measures side by side. The test shows that take rates moved, but not what the platforms changed to move them.*

AFTER (~~deleted~~, added)

> **The thin take rate of Indonesian and regional platforms explains much of the threefold difference, an exploratory finding. Measured in percentage points, both groups changed their take rates by about 1 point a year, and the difference between them is nowhere near large enough to count (p = 0.65).** *The two groups differ in the take rate they start from. The typical Indonesian and regional platform keeps about 1.6 percent of each sale, against about 17 percent abroad. The thinnest take rates belong to the three marketplaces, while Grab and GoTo keep shares closer to those abroad.*

As Section 3.2 showed, the same 1-point rise then has very different effects. It lifts revenue by more than half at the thinner take rate and by about 6 percent at the thicker one. Figure 5.1 shows the three measures side by ~~side.~~ **side, in its Panels A, B and C.** *The test shows that take rates moved, but not what the platforms changed to move them.*

BEFORE

Figure 5.1. The thin take rate. *Each dot is one platform's median over its years. Left: the share of each sale that the platform keeps, on a log scale. Middle: the yearly change in that share, in percentage points. Right: the same change relative to the share itself, as a log change in percent. Black bars mark group medians. The foreign group has 26 platforms here and 27 in the main test, because one platform has no pair of usable years for the change in points. Source: company filings; author's calculations. Exploratory.*

AFTER (~~deleted~~, added)

Figure 5.1. The thin take rate. *Each dot is one platform's median over its years. ~~Left:~~* **Panel A:** *the share of each sale that the platform keeps, on a log scale. ~~Middle:~~* **Panel B:** *the yearly change in that share, in percentage points. ~~Right:~~* **Panel C:** *the same change relative to the share itself, as a ~~log change in percent.~~* **simple percentage change.** *Black bars mark group medians. The foreign group has 26 platforms here and 27 in the main test, because one platform has no pair of usable years for the change in points. Source: company filings; author's calculations. Exploratory.*

5.3 What explains the wedge and the ecosystem ratio (was: 5.3 What moved the take rate)

NEW

Size explains the wedge, and pricing explains the ecosystem ratio. Table 5.4 regresses each on the features of the platform and of its home market.

NEW

Table 5.4. What explains the wedge and the ecosystem ratio

NEW

This table reports OLS estimates of equation (7). The dependent variable is the natural logarithm of the wedge in US dollars in columns (1) to (3) and the ecosystem ratio, capped at its 5th and 95th percentiles, in columns (4) to (6). IndonesiaRegion equals one for the Indonesian and regional platforms. Size is total assets in US dollars. The base type is marketplace. Columns (2), (3), (5) and (6) add the GDP growth, inflation, household spending growth and policy rate of the platform's main market, and columns (3) and (6) add year dummies. Appendix A defines the variables. Standard errors (in parentheses) are clustered at the platform level. ∗∗∗, ∗∗ and ∗ correspond to statistical significance at the 1%, 5% and 10% levels. The last row gives the p-value of a wild cluster bootstrap with 2,999 replications for IndonesiaRegion.

NEW

Dependent variable	*ln(Wedge)*	*ln(Wedge)*	*ln(Wedge)*	*Ecosystem ratio*	*Ecosystem ratio*	*Ecosystem ratio*
	(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*

… (table of 30 rows)

NEW

Note: The take rate and its parts are not regressors, because the ecosystem ratio is one divided by the take rate, minus one. Source: company filings; filings with the US Securities and Exchange Commission; World Bank; Bank for International Settlements; author's calculations. Exploratory.

NEW

The wedge scales with the size of the platform. A 1 percent rise in total assets goes with a rise of about 0.9 percent in the wedge, and the regressors of column (1) explain 81 percent of its variation. Once size is held fixed, Indonesian and regional platforms do not carry a different wedge. The coefficient on IndonesiaRegion is −0.07 in column (1) and stays between −0.30 and 0.08 in the others, never close to significance. On-demand platforms carry about 60 percent less wedge for their assets than marketplaces do.

NEW

The ecosystem ratio is different. The Indonesian and regional platforms have a ratio 13 to 17 points above foreign platforms of the same type, size and age, in markets with the same growth, inflation and interest rates (columns 4 to 6). The foreign mean is about 9. The wild-bootstrap p-values, which allow for the small number of clusters, are between 0.005 and 0.04. Size and age explain little of the ratio. Type explains a good deal: on-demand platforms have ratios 17 to 20 points below marketplaces, which means that they keep a much larger share of each sale, and travel, freelance and resale platforms are 7 to 14 points below. A home market that grows faster has a higher ratio, by about 1.3 to 1.5 points for each point of GDP growth, an exploratory pattern that the thesis does not read as causal.

NEW

The result for the Indonesian and regional platforms survives the checks of Appendix C. The coefficient stays between 15 and 18 when each of the five platforms is dropped in turn, and it is 21 without Alibaba, Pinduoduo, Rakuten and Shopify. It is 28.5 when the ratio is not capped (p = 0.008), 17 when a dummy for emerging home markets is added, whose own coefficient is close to zero (0.4, p = 0.94), and 17 without the six platforms whose main market is ambiguous. So the thin take rates of Indonesian and regional platforms are not a general emerging-market pattern. Total assets are those of the whole company, which is larger than the platform business at GoTo, Rakuten and Sea, so size is measured with error, and the result rests on five platforms.

REMOVED

Pricing, not more commerce, moved revenue at the Indonesian and regional platforms. Pricing here means a change in the take rate, through fees or incentives. With equation (5), the take rate accounts for about two thirds of a revenue move at the typical pair, against about a quarter abroad. The wedge, by contrast, follows the commerce: in both groups, about 95 percent of a move in the wedge comes from transaction value.

REMOVED

Where companies disclose both their fees and their incentives, equation (6) splits the growth of revenue further. Table 5.4 gives the seven traced windows. Fees and incentives are measured per unit of transaction value, and a negative share means that a part pulled revenue down. The windows of one platform overlap, so they are not seven independent cases.

REMOVED

Table 5.4. What moved revenue in seven Indonesian and regional windows (share of the growth of revenue, percent)

REMOVED

Platform and window	*Growth of revenue*	*More commerce*	*Higher fees*	*Lower incentives*
Tokopedia, 2022–23	*+53%*	*−22*	*76*	*45*

… (table of 7 rows)

REMOVED

Note: Shares are rounded, so a row can add to 99 or 101. Source: company filings; author's calculations (Appendix A).

BEFORE

Higher fees and lower incentives together outweighed more commerce in every window. Lower incentives were the larger of the two in four windows and higher fees in two. In the seventh, Grab from 2021 to 2024, the two were nearly equal.

AFTER (~~deleted~~, added)

The regression does not say how a platform moved its take rate. Company reports do, for the seven windows traced in Appendix B, where the growth of revenue is split into more commerce, higher fees and lower incentives. *Higher fees and lower incentives together outweighed more commerce in every window. Lower incentives were the larger of the two in four windows and higher fees in two. In the seventh, Grab from 2021 to 2024, the two were nearly equal.* **The windows of one platform overlap, so they are not seven independent cases. Blibli does not report incentives, so its split rests on a proxy, and part of its 2022–23 rise comes from travel and digital products. Bukalapak's reports do not allow the parts to be separated.**

REMOVED

Three series need a note. GoTo's 2022 transaction value is an estimate. Blibli does not report incentives, so its split rests on a proxy, and part of its 2022–23 rise comes from travel and digital products. Bukalapak's reports do not allow the parts to be separated.

BEFORE

Tokopedia can be read in two ways, and both point to pricing. Its incentives fell by about a third in 2023. Counted line by line in its accounts, lower incentives gave about 60 percent of the extra revenue and higher gross fees about 40 percent. That count does not adjust for the fall in transaction value, which lowered both lines. Measured per unit of transaction value, as in Table 5.4, higher fees account for more of the rise in the take rate. The two views differ only on which lever led (Figure 5.2).

AFTER (~~deleted~~, added)

Tokopedia can be read in two ways, and both point to pricing. Its incentives fell by about a third in 2023. Counted line by line in its accounts, lower incentives gave about 60 percent of the extra revenue and higher gross fees about 40 percent. That count does not adjust for the fall in transaction value, which lowered both lines. Measured per unit of transaction value, ~~as in Table 5.4,~~ higher fees account for more of the rise in the take rate. The two views differ only on which lever led (Figure 5.2).

NEW

The ecosystem ratio matters for three groups. For investors, a high ratio means that small changes in fees or discounts produce large swings in revenue growth. At the typical Indonesian and regional ratio of 35, the platform keeps about 2.8 percent of each sale, and a rise of 1 point in the take rate lifts revenue by about 36 percent at constant volume. At the typical foreign ratio of 5.5, the same rise lifts it by 6.5 percent. So a revenue multiple means something different in the two groups, because each unit of revenue stands for 36 units of transaction value in one and 6.5 in the other. Growth that comes from the take rate cannot go on without end, and Section 5.4 shows that markets priced transaction value.

NEW

For sellers and drivers, a high ratio means that most of each sale goes to them and not to the platform. A rising take rate means that the platform keeps more of each sale. The take rate rose in 87 percent of the Indonesian and regional pairs, against 65 percent abroad, and at Tokopedia part of the rise came from smaller discounts, which are paid to buyers. Whether that shows market power is a question the thesis cannot answer, because it holds no data on what sellers and drivers paid or could have chosen instead. The pattern is also consistent with platforms that listed in 2021 and 2022 moving from growth to earnings, since their largest moves came one or two years after listing.

NEW

For policy, the wedge is the base that revenue-based rules do not reach. A rule that acts on platform revenue acts on about 3 percent of the commerce at a typical Indonesian or regional platform. A cap on commissions acts on the fee rate, and a limit on discounts

acts on the incentive rate, the two parts of the take rate in equation (5). Chapter 6 sets each rule against the part it reaches.

5.4 What markets valued

BEFORE

Across 28 listed platforms and 138 platform-years, market value moved with transaction value, a registered result. A 1 percent rise in transaction value goes with a rise of about 0.6 percent in market value (β_V in equation (8), p = 0.002). Section 4.5 describes this sample, which differs from the 27 foreign platforms of H1.

AFTER (~~deleted~~, added)

Across 28 listed platforms and 138 platform-years, market value moved with transaction value, a registered result. A 1 percent rise in transaction value goes with a rise of about 0.6 percent in market value (β_V in equation ~~(8),~~ **(9),** *p = 0.002). Section 4.5 describes this sample, which differs from the 27 foreign platforms of H1.*

5.5 The platforms and the sellers in 2023

BEFORE

Figure 5.3. Growth in 2023 (percent). *Blue bars show measures of buying; orange bars show the figures that described a boom. Each measure is in its own unit. Sources: company reports, Bank Indonesia, Momentum Works and BPS; household spending from national accounts. GoTo's change rests on an estimated 2022 value, and Momentum Works' 3.5 is the author's calculation from its published levels (its 2024 release states 3.7). Appendix D gives the values.*

AFTER (~~deleted~~, added)

Figure 5.3. Growth in 2023 (percent). *Blue bars show measures of buying; orange bars show the figures that described a boom. Each measure is in its own unit. Sources: company reports, Bank Indonesia, Momentum Works and BPS; household spending from national accounts. GoTo's change rests on an estimated 2022 value, and Momentum Works' 3.5 is the author's calculation from its published levels (its 2024 release states 3.7). Appendix ~~D~~* **E** *gives the values.*

5.6 H2 and the entry test

BEFORE

H2 holds on BPS's published figures, but the registered entry test shows that first-time entry cannot explain the whole rise in the count. From 2022 to 2023, online sales value grew by 41 percent and the number of businesses by 27 percent. More businesses account for 71 percent of that growth, and higher sales per business for the rest.

AFTER (~~deleted~~, added)

H2 holds on BPS's published figures, but the registered entry test shows that first-time entry cannot explain the whole rise in the count. From 2022 to 2023, online sales value grew by 41 ~~percent and the~~ **percent. The** *number of businesses* **grew** *by 27 ~~percent. More~~*

percent and sales per business by 10 percent, so more *businesses account for ~~71 percent~~* **most** *of ~~that growth, and higher sales per business for~~ the ~~rest.~~* **growth.**

BEFORE

H2 holds again from 2023 to 2024. Value grew by 17 percent, and more businesses account for 90 percent of the growth, or 71 percent on BPS's other count for 2023. The split of equation (7) is arithmetic, and it cannot tell a new seller from one counted for the first time.

AFTER (~~deleted~~, added)

H2 holds again from 2023 to 2024. Value grew by 17 percent, **the number of businesses by 15 percent** *and ~~more businesses account for 90 percent of the growth,~~* **sales per business by 1.5 percent,** *or ~~71~~* **by 12 and 4.7** *percent on BPS's other count for 2023. The split of equation ~~(7)~~* **(8)** *is arithmetic, and it cannot tell a new seller from one counted for the first time.*

5.7 What the rise in the count contains

NEW

This table splits the rise in the number of online sellers between the weighted totals of the two BPS survey files into net entry and earlier sellers who were newly counted. The first column assumes that no seller stopped between the surveys, and the second assumes that sellers stopped at 5 percent a year. The percentages are shares of the rise.

BEFORE

H2 holds on the published series, but it sits right on its threshold once the extra count is read as coverage, an exploratory follow-up. H2's split treats every seller in the count as having entered. Suppose instead that the sellers not explained by entry were reached by wider coverage, and that their sales came with them. With both taken out of the 2023 figures, more businesses account for about 54 percent of the growth if no seller stopped, and about 42 percent if sellers stopped at 5 percent a year. H2 requires more than 50 percent. Taking out the sellers but keeping their sales gives 34 to 47 percent, but that version treats the count and the value unevenly.

AFTER (~~deleted~~, added)

H2 holds on the published series, but it sits right on its threshold once the extra count is read as coverage, an exploratory follow-up. H2's split treats every seller in the count as having entered. Suppose instead that the sellers not explained by entry were reached by wider coverage, and that their sales came with them. With both taken out of the 2023 figures, ~~more~~ **the number of** *businesses ~~account for about 54~~* **grows by 17** *percent ~~of the growth~~ if no seller stopped, ~~and about 42~~* **against 14 to 15** *percent ~~if~~* **for sales per business, so H2 holds narrowly. If** *sellers stopped at 5 percent a ~~year.~~* **year, the number of businesses grows by 12 percent and sales per business by about 16 to 17 percent, so H2 fails by a few points.** *H2 requires ~~more than 50 percent.~~* **the first**

growth rate to be the larger. *Taking out the sellers but keeping their sales gives ~~34 to 47~~* **17 against 20 percent, and 12 against 25** *percent, but that version treats the count and the value unevenly.*

5.10 Why the national figures disagree

BEFORE

> *In 2024 the figures divide by coverage. Those that cover mainly platforms grew by 5 to 7 percent, and BPS's grew by about 17 percent. The adjacent figures of Table 4.3 point the same way as BPS but do not measure Indonesian e-commerce. Parcels shipped in Southeast Asia grew by about 25 percent, and the tax on foreign digital services by about 25 percent.*

AFTER (~~deleted~~, added)

> *In 2024 the figures divide by coverage. Those that cover mainly platforms grew by 5 to 7 percent, and BPS's grew by about 17 percent. The adjacent figures of ~~Table 4.3~~* **Appendix A** *point the same way as BPS but do not measure Indonesian e-commerce. Parcels shipped in Southeast Asia grew by about 25 percent, and the tax on foreign digital services by about 25 percent.*

BEFORE

> *Payment figures grew faster than any of these from 2023 to 2024. E-money shopping grew by about 30 percent, mobile-banking purchases by 83 percent and QRIS payments by 187 percent. They are context only, because they include in-person payments and the switch away from cash.*

AFTER (~~deleted~~, added)

> *Payment figures grew faster than any of these from 2023 to 2024. E-money shopping grew by about 30 percent, mobile-banking purchases by 83 percent and ~~QRIS~~ payments* **through the Quick Response Code Indonesian Standard (QRIS)** *by 187 percent. They are context only, because they include in-person payments and the switch away from cash.*

Chapter 6. Discussion and policy implications

6.1 The rules and what each acts on

NEW

> *This table lists each recent Indonesian rule that bears on the figures in Chapters 4 and 5. For each rule it gives what the rule does, which part of the figures it acts on, the section of Chapter 5 that measures that part, and the status of the legal text when this thesis was written. Amounts in rupiah (Rp) are converted to US dollars in the note.*

BEFORE

Rule	*What it does*	*What it acts on*	*Measured in*	*Status of the text*
Commission cap (Transport Minister Decree KP-PHB 532/2026; Presidential Regulation 27/2026)	*Caps the commission on two-wheel passenger trips at 8 percent from 1 July 2026*	*The fee rate, for one service*	*5.3*	*Decree in force from July 2026; presidential regulation released 8 October 2026; neither text public when this thesis was written*

… (table of 7 rows)

AFTER (~~deleted~~, added)

Rule	*What it does*	*What it acts on*	*Measured in*	*Status of the text*
Commission cap (Transport Minister Decree KP-PHB 532/2026; Presidential Regulation 27/2026)	*Caps the commission on two-wheel passenger trips at 8 percent from 1 July 2026*	*The fee rate, for one service*	*5.3*	*Decree in force from July 2026; presidential regulation released 8 October 2026; neither text public when this thesis was written*

… (table of 7 rows)

BEFORE

Equation (6) shows where the first two rules act. The take rate is the fee rate less the incentive rate. A cap on a fee pushes the take rate down. A limit on incentives pushes it up. The two rules act on opposite sides of the same take rate.

AFTER (~~deleted~~, added)

Equation ~~(6)~~ **(5)** *shows where the first two rules act. The take rate is the fee rate less the incentive rate. A cap on a fee pushes the take rate down. A limit on incentives pushes it up. The two rules act on opposite sides of the same take rate.*

6.2 What the commission cap and the discount rule can move

BEFORE

The discount rule acts on the lever that moved take rates most often. Lower incentives were the larger part of the rise in the take rate in four of the seven windows of Table 5.4. The rule binds every online trade operator, including ride-hailing apps that sell food or goods. Whether it reaches ride fares, which the transport ministry sets, is unclear.

AFTER (~~deleted~~, added)

The discount rule acts on the lever that moved take rates most often. Lower incentives were the larger part of the rise in the take rate in four of the seven windows of ~~Table 5.4.~~ **Appendix B.** *The rule binds every online trade operator, including ride-hailing apps that sell food or goods. Whether it reaches ride fares, which the transport ministry sets, is unclear.*

6.3 How far the marketplace tax reaches

BEFORE

> *The record reaches further than the tax. Every domestic seller on a designated marketplace must give the marketplace a tax number or national ID and an address before being paid. Exempt sellers file a statement as well. Tax research finds that records held by a third party improve compliance where they exist (Section 2.3).*

AFTER (~~deleted~~, added)

> *The record reaches further than the tax. Every domestic seller on a designated marketplace must give the marketplace a tax number or national ~~ID~~* **identity number** *and an address before being paid. Exempt sellers file a statement as well. Tax research finds that records held by a third party improve compliance where they exist (Section 2.3).*

6.5 What the registered predictions test

NEW

> *This table lists the registered predictions on the 2026 rules. For each prediction it gives what is expected, the rule it bears on and the date it is scored. The outcome stays blank until the company or BPS releases the figure.*

BEFORE

> *Note: Ranges as registered; registration dates are in Appendix C. A tenth prediction, that the start of the marketplace tax would slip again, is withdrawn: collection had begun on 1 October 2026, a day before the prediction was registered. Sources for scoring: company results releases; BPS national accounts and e-commerce publication.*

AFTER (~~deleted~~, added)

> *Note: Ranges as registered; registration dates are in Appendix ~~C.~~* **D.** *A tenth prediction, that the start of the marketplace tax would slip again, is withdrawn: collection had begun on 1 October 2026, a day before the prediction was registered. Sources for scoring: company results releases; BPS national accounts and e-commerce publication.*

BEFORE

> **[Results slot]** *To be written after the GoTo results (late October), the BPS national accounts (early November) and the Grab and Sea results (mid-November). For each outcome: the result against its range in Table 6.2 and Appendix C, then one paragraph on what the outcomes show about the cap and one on marketplace fees.*

AFTER (~~deleted~~, added)

> **[Results slot]** *To be written after the GoTo results (late October), the BPS national accounts (early November) and the Grab and Sea results (mid-November). For each outcome: the result against its range in Table 6.2 and*

Appendix ~~C,~~ **D,** *then one paragraph on what the outcomes show about the cap and one on marketplace fees.*

6.6 Beyond Indonesia

BEFORE

> *Commission caps have been tried elsewhere as well. Fourteen US cities and states capped the commissions that delivery platforms charged independent restaurants. The platforms raised delivery fees for consumers and showed chain restaurants first more often. Independent restaurants lost orders and revenue, while chains gained (Li and Wang 2025). In the terms of equation (6), a cap on one fee left the platform free to change another. Indonesia's cap also covers one fee on one service.*

AFTER (~~deleted~~, added)

> *Commission caps have been tried elsewhere as well. Fourteen US cities and states capped the commissions that delivery platforms charged independent restaurants. The platforms raised delivery fees for consumers and showed chain restaurants first more often. Independent restaurants lost orders and revenue, while chains gained (Li and Wang 2025). In the terms of equation ~~(6),~~* **(5),** *a cap on one fee left the platform free to change another. Indonesia's cap also covers one fee on one service.*

Chapter 7. Conclusion

7.1 The answer to the research question

BEFORE

> *The answer to the second half differs by link. On the inner link, the choice of revenue changed the conclusion, above all from 2022 to 2024. The Indonesian and regional platforms changed their take rates about three times as much as platforms abroad, measured against the take rate itself, so H1 is rejected. They keep a much thinner share of each sale, so price changes of a similar size in points moved their revenue far more. In the windows traced, pricing moved revenue more than commerce did. Market value, for its part, moved with transaction value, and the data do not show that markets valued growth from pricing any less.*

AFTER (~~deleted~~, added)

> *The answer to the second half differs by link. On the inner link, the choice of revenue changed the conclusion, above all from 2022 to 2024.* **Abroad, revenue growth matched transaction value growth one for one, with a slope of 0.99, so revenue was a workable stand-in. In Indonesia and the region the slope was 0.40, and revenue grew by about 44 percent a year whatever transaction value did, so H1 is rejected there.** *The Indonesian and regional platforms changed their take rates about ~~three~~* **3.6** *times as much as platforms ~~abroad, measured against the take rate itself, so H1 is rejected.~~* **abroad.** *They* **also** *keep a much thinner share of each ~~sale, so~~* **sale: their ecosystem ratio is 13 to 17 points above that of comparable foreign platforms, after the type, size and age of the platform and the conditions of its home market are held fixed. So** *price changes of a similar size in points moved their revenue far more.*

In the windows traced, pricing moved revenue more than commerce did. Market value, for its part, moved with transaction value, and the data do not show that markets valued growth from pricing any less.

BEFORE

On the outer link, H2 holds on BPS's published figures: more businesses account for most of the growth in online sales value. The registered entry test failed, and about half of the 2023 rise in the count is not explained by first-time entry. The extra count leaned toward long-running businesses. Read as wider coverage, an exploratory check, the extra count leaves H2 on its threshold: more businesses then account for about 42 to 54 percent of the growth, against the 50 percent H2 requires.

AFTER (~~deleted~~, added)

On the outer link, H2 holds on BPS's published figures: more businesses account for most of the growth in online sales value. The registered entry test failed, and about half of the 2023 rise in the count is not explained by first-time entry. The extra count leaned toward long-running businesses. Read as wider coverage, an exploratory check, the extra count leaves H2 ~~on~~ **near** *its threshold: ~~more~~* **the number of** *businesses then ~~account for about 42~~* **grows by 12** *to ~~54~~* **17** *percent ~~of~~* **and sales per business by 14 to 17 percent, so H2 holds narrowly in one reading and fails by a few points in** *the ~~growth, against the 50 percent H2 requires.~~* **other.**

7.2 What the thesis adds

BEFORE

For investors, the thesis adds a check that can be run on any platform that reports both amounts. Revenue growth splits into more commerce and a larger take rate with two public figures (equation (5)). At the Indonesian and regional platforms, the take rate accounted for about two thirds of a revenue move at the typical pair. Abroad it accounted for about a quarter. The thinner the take rate, the more a small change in fees or incentives moves revenue growth. Across 28 listed platforms, market value moved with transaction value, as a registered prediction stated. A 1 percent rise in transaction value went with about 0.6 percent more market value.

AFTER (~~deleted~~, added)

For investors, the thesis adds a check that can be run on any platform that reports both amounts. Revenue growth splits into ~~more commerce~~ **growth in transaction value** *and a ~~larger~~* **change in the** *take rate with two public figures (equation ~~(5)).~~* **(4)).** *At the Indonesian and regional platforms, ~~the take rate accounted~~* **revenue growth moved little with transaction value, while abroad it moved one** *for ~~about two thirds of a revenue move at the typical pair. Abroad it accounted for about a quarter.~~* **one.** *The thinner the take rate, the more a small change in fees or incentives moves revenue growth.* **At the typical Indonesian and regional ecosystem ratio of 35, a rise of 1 point in the take rate lifts revenue by about 36 percent at constant volume, against 6.5 percent at the typical foreign ratio of 5.5. A revenue multiple then stands for 36 units of transaction value per unit of revenue in one group and 6.5**

in the other. *Across 28 listed platforms, market value moved with transaction value, as a registered prediction stated. A 1 percent rise in transaction value went with about 0.6 percent more market value.*

7.3 Predictions that failed

BEFORE

Appendix C lists every registered test with its outcome. The predictions on the 2026 rules are listed in Section 6.5, with the date on which each is scored. One of them, that the start of the marketplace tax would slip again, was withdrawn, because the tax had started a day before the prediction was registered. A comparison of seven countries on firms' digital records was withdrawn as a test for a similar reason. Published figures seen at registration already settled it, so it is reported as descriptive.

AFTER (~~deleted~~, added)

Appendix ~~C~~ **D** *lists every registered test with its outcome. The predictions on the 2026 rules are listed in Section 6.5, with the date on which each is scored. One of them, that the start of the marketplace tax would slip again, was withdrawn, because the tax had started a day before the prediction was registered. A comparison of seven countries on firms' digital records was withdrawn as a test for a similar reason. Published figures seen at registration already settled it, so it is reported as descriptive.*

7.4 Limits of the evidence

BEFORE

The invisible wedge is the part of each sale that a platform passes on to sellers, drivers and others. It is not missing output, unrecorded income or unpaid tax. The thesis does not find BPS's figures wrong in general, nor that total online buying in Indonesia fell. It finds that one year's count of sellers moved for reasons other than first-time entry, and that the figures disagree where they cover different sellers. The splits of Table 5.4 are accounting, not causes. Marketplace sellers keep financial statements more often, but the thesis does not show that marketplaces lead them to.

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BEFORE

The tests of H1 are one-sided, because each asks whether the Indonesian and regional group moves more. A two-sided test would about double each p-value, and the main Mann–Whitney result would still be below 1 percent. The result depends on the larger

movers. Dropping the two largest, Blibli and Tokopedia, leaves p = 0.058, and keeping only the first five years after each platform's listing leaves p = 0.059 (Appendix B).

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NEW

The new regressions are exploratory and rest on few platforms. The growth test for Indonesia and the region has 15 pairs from five platforms, and the Indonesia coefficients of Table 5.4 are identified by the same five. A wild cluster bootstrap allows for the small number of clusters but cannot remove the limit. Total assets are those of the whole company, so size is overstated at GoTo, Rakuten and Sea. The home market of each platform is assigned by the author, and the policy rate is missing for Kazakhstan and Nigeria. The regressions describe how the wedge and the ecosystem ratio differ across platforms and do not show what causes the difference.

BEFORE

Foreign platforms also differ in how they book shopper discounts. Eight of the 27 record them mainly as marketing costs, not as lower revenue, which could steady their take rates. Their take rates were not steadier, and without them the test of H1 still gives p = 0.004, an exploratory check (Appendix B).

AFTER (~~deleted~~, added)

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D. My review notes

Things I would not claim more strongly than the text does

1. The Indonesia and region result rests on five platforms. The wild bootstrap allows for the small number of clusters, but it does not create information. In the growth test the Indonesian and regional group has 15 pairs, and the slope's p-value (0.02) uses a t distribution with four degrees of freedom. I removed a claim that the two correlations differ at p < 0.001 for the same reason.
2. Total assets are for the whole company. Tokopedia and GoTo (on-demand) carry the same figure, the GoTo group's. Rakuten's includes its bank.
3. I assigned each platform's main market and type. Six assignments are ambiguous; the result is unchanged without them (Appendix C.3).

4. The size and macro controls explain the wedge, not the difference between the groups: the wedge is almost all transaction value, so the wedge regression mostly shows that big platforms carry a lot. The ecosystem ratio is where the groups differ. Prof. Kong may ask why the wedge regression is there; the honest answer is that she asked for it, and that its result is itself informative.
5. "Market power" in the sellers and drivers paragraph is stated as an open question, because the thesis has no seller-side data.

Where the text got longer, against her request to tighten. The new tests add about 2,070 words net. Candidates to cut or move to Appendix C, with the current word counts: the paragraphs on the narrower first test, the direction of moves, the quarterly pairs and age in 5.2 (296 words); the account of the seven windows and the company sources in 4.4 (217 words); 7.4 on the limits (942 words) overlaps with the new limits paragraph. Section 5.2 is now 1896 words and 5.3 is 1688.

Inconsistencies I know about. Figure 5.2, the Tokopedia split and Appendix B.3 are in logs because the split of revenue growth into commerce, fees and incentives is exact only in logs; Appendix C.1 keeps the first version of the H1 test in logs. The market-value regression keeps its registered log definition. A simple-change version needs the licensed market values and a local run, which I have not written.

Decisions I need from you

1. Send Prof. Kong the nine questions in `KONG_COMMENTS_TRACE.md`, section 6, or only the ones still open: the unit of the H2 regression, whether Tables 13 and 14 (now 6.1 and 6.2) and the older figures stay, what counts as competitive entry or a regulatory event for the foreign platforms, and whether the entry test belongs inside the further analyses.
2. What to cut for length. The layout step is done, so cuts no longer move it, but each cut changes the page numbers in the YZU file (rebuilt in one command).
3. Whether the abstract (195 words) stays as it is; the new second paragraph adds four words.

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1. The files hold 11,848, 31,724 and 39,736 rows. Each weighted total is within 0.1 percent of the BPS count: 2,361,423; 2,995,879, implied by the 2023 count and its growth of 27.4 percent; and 3,816,750.↩